Extended Appendix

How Do We Hear a Rāga? An Interpretable Cognitive Model of Melodic Recognition

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This appendix supplements the 10-page project report with mathematical definitions, per-rāga feature visualisations, all confusion matrices, and experimental results. All figures are generated by the pipeline scripts in the GitLab repository.

1 Mathematical Definitions

This section defines every mathematical operation used in the pipeline.

1.1 Cents Conversion

The cent is a logarithmic unit of pitch interval where one octave equals 1200 cents. Given a fundamental frequency f_0 and tonic frequency Sa_{Hz}:

$$c = 1200 \times \log_2\left(\frac{f_0}{\text{Sa}_{\text{Hz}}}\right)$$

Example: If Sa = 150 Hz and $f_0 = 225$ Hz, then $c = 1200 \times \log_2(1.5) = 1200 \times 0.585 = 702$ cents (Pa, the perfect fifth). The logarithmic scale ensures that equal cent intervals sound perceptually equal regardless of register.

1.2 Octave Folding

All pitches are folded into a single octave $[0, 1200)$ using modular arithmetic:

$$c_{\text{mod}} = c \bmod 1200$$

This implements octave equivalence: Sa at 150 Hz and Sa at 300 Hz both map to 0 cents.

1.3 Circular Distance (Swara Mapping)

The 12 swara bins are arranged on a circle. The distance from a pitch c to bin b is:

$$d(c, b) = \min\big(|c - b|,\ 1200 - |c - b|\big)$$

This handles the octave boundary correctly: a pitch at 1190 cents is 10 cents from Sa (0/1200), not 1190 cents away. If $d(c, b) \leq 40$ cents, the frame is assigned swara b; otherwise it is labelled "glide."

1.4 Harmonic Tonic Scoring

For each tonic candidate frequency, a pitch-class histogram h is built (120 bins, 10-cent resolution). The score is:

$$\text{score} = \sum_{|i|<30} h[i] \ + \ 0.5 \times \sum_{|i-702|<30} h[i]$$

where the first term is Sa mass (0 cents) and the second is Pa mass (702 cents). The 0.5 weight reflects the tānpura drone structure: Sa is the primary tone, Pa is secondary reinforcement.

1.5 Weighted F1 Score

$$F_{1,\text{weighted}} = \sum_{i=1}^{K} \frac{n_i}{N} \times F_{1,i}$$

where n_i is the number of samples in class i, N is total samples, and $K = 13$ classes. Chosen over accuracy (which rewards ignoring minority classes) and macro F1 (unstable with small classes).

1.6 Permutation Test (H4)

The observed within/between-family confusion ratio is:

$$r_{\text{obs}} = \frac{\overline{C}_{\text{within}}}{\overline{C}_{\text{between}}} = 3.27$$

For $n = 10{,}000$ iterations, family labels are randomly shuffled and r_{perm} recomputed. The p-value is the fraction of permutations where $r_{\text{perm}} \geq r_{\text{obs}}$. Result: $p = 0.0004$.

2 Pipeline Visualisation

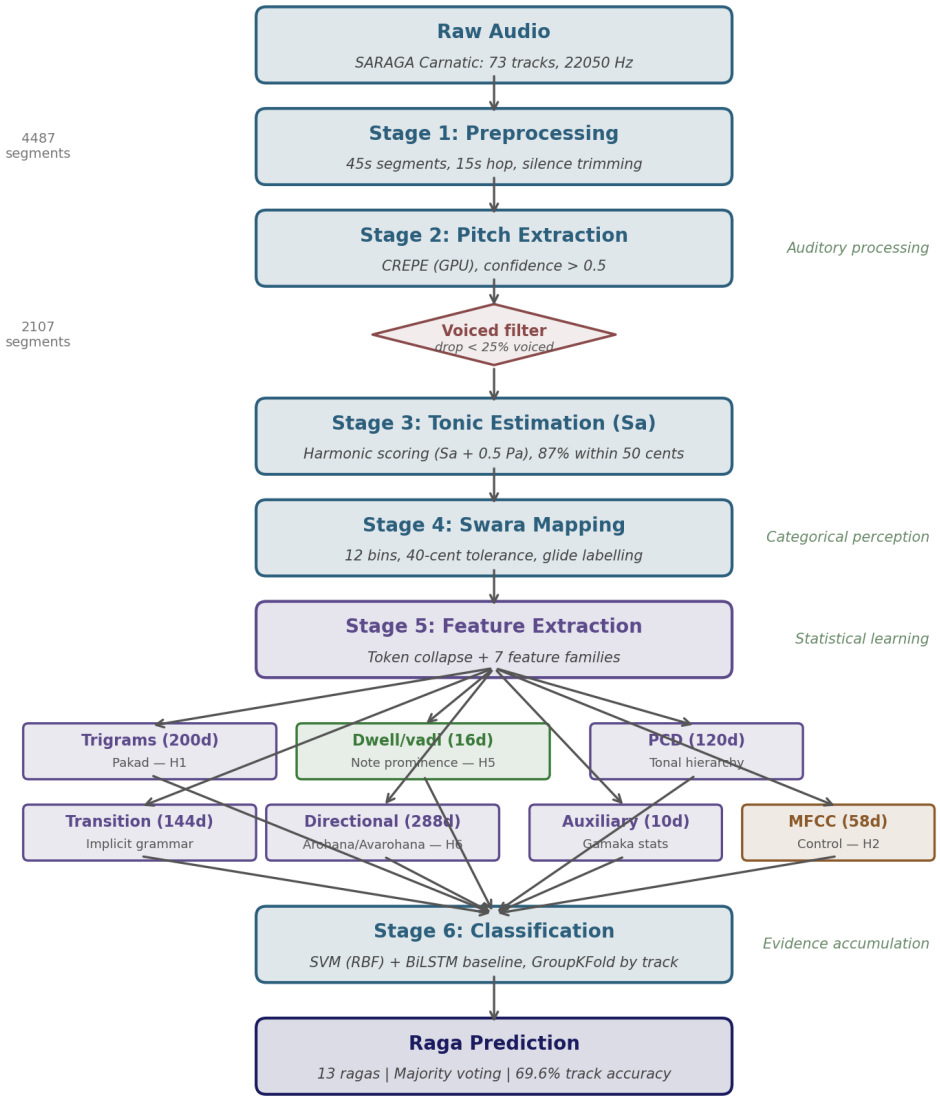
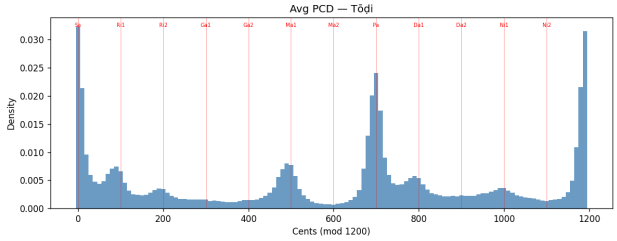


Figure 1: Six-stage pipeline architecture. Each stage mirrors a cognitive or musicological process. The seven feature families feed independently into the SVM classifier; Dwell/vādi (16d) achieves the best track accuracy.

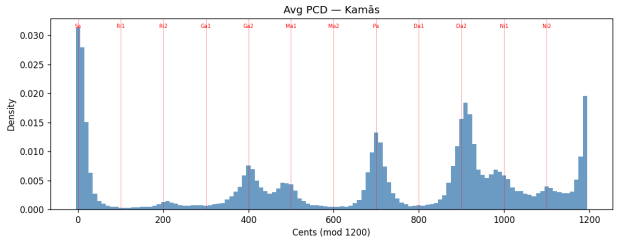
3 Per-Rāga Feature Visualisations

These figures show per-rāga pitch class distributions (120-bin histograms), swara transition matrices (12×12 heatmaps), and directional features. They demonstrate that rāgas with distinctive profiles classify well, while Harikāmbhōji-family rāgas with overlapping profiles are confused.

3.1 Pitch Class Distributions

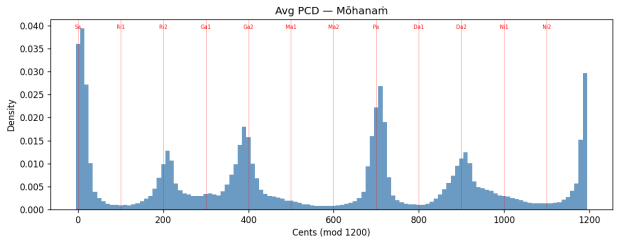


(a) Tōḍi (0.86 F1) — unique intervallic signature

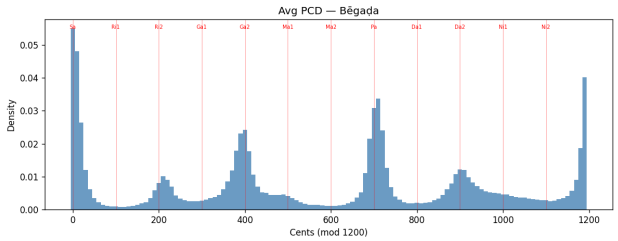


(b) Kamās (0.36 F1) — overlaps with Harikāmbhōji family

Figure 2: Pitch class distributions. Tōḍi's distinctive Ri1–Ga1 peaks explain its high classification performance. Kamās's PCD is nearly identical to its parent melakarta, explaining the family confusion.



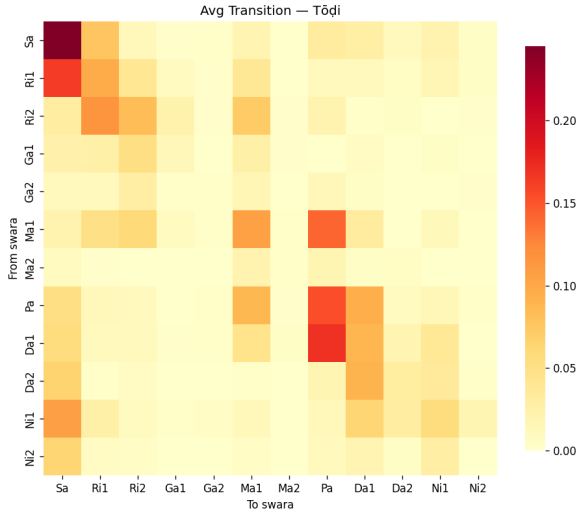
(a) Mōhanaṁ (pentatonic)



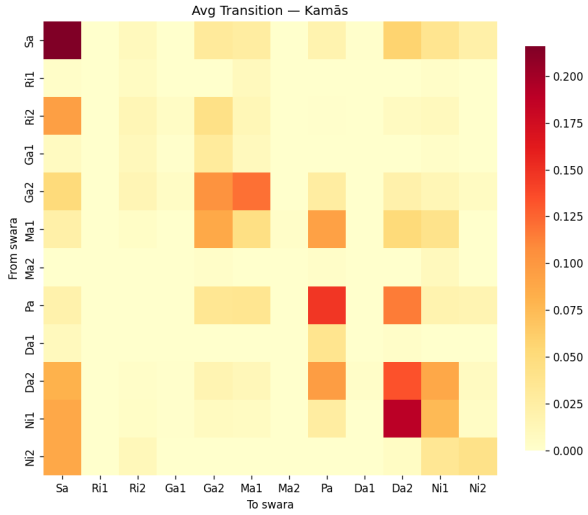
(b) Bēgaḍa

Figure 3: Additional PCDs. Mōhanaṁ's pentatonic profile (5 peaks, no Ma or Ni) is distinctive but every note also appears in the Harikāmbhōji family, making it hard to classify by presence alone.

3.2 Transition Matrices

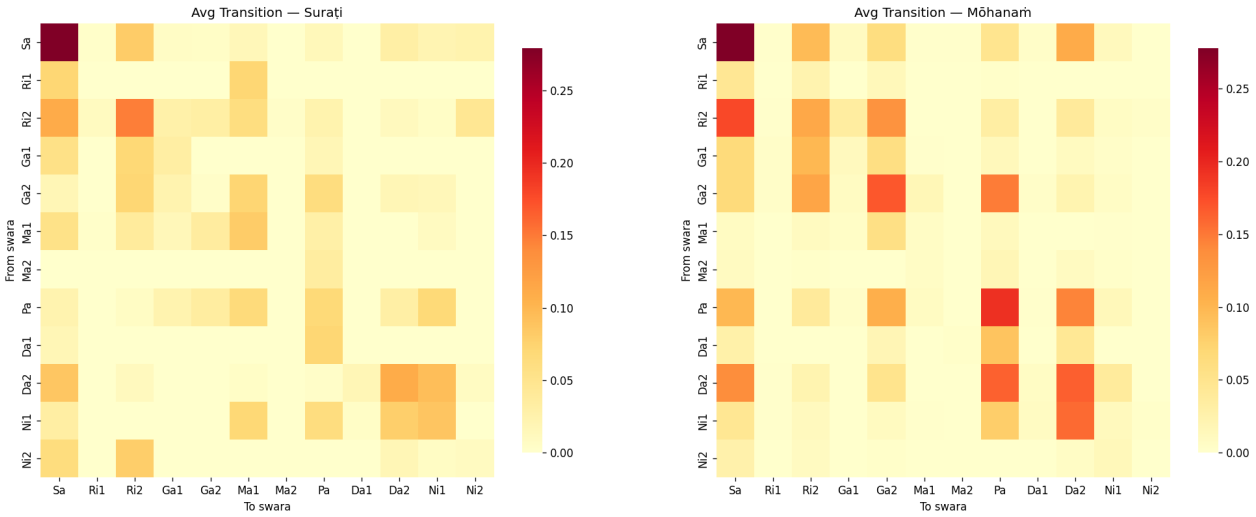


(a) Tōḍi — concentrated transitions



(b) Kamās — diffuse, shared with family

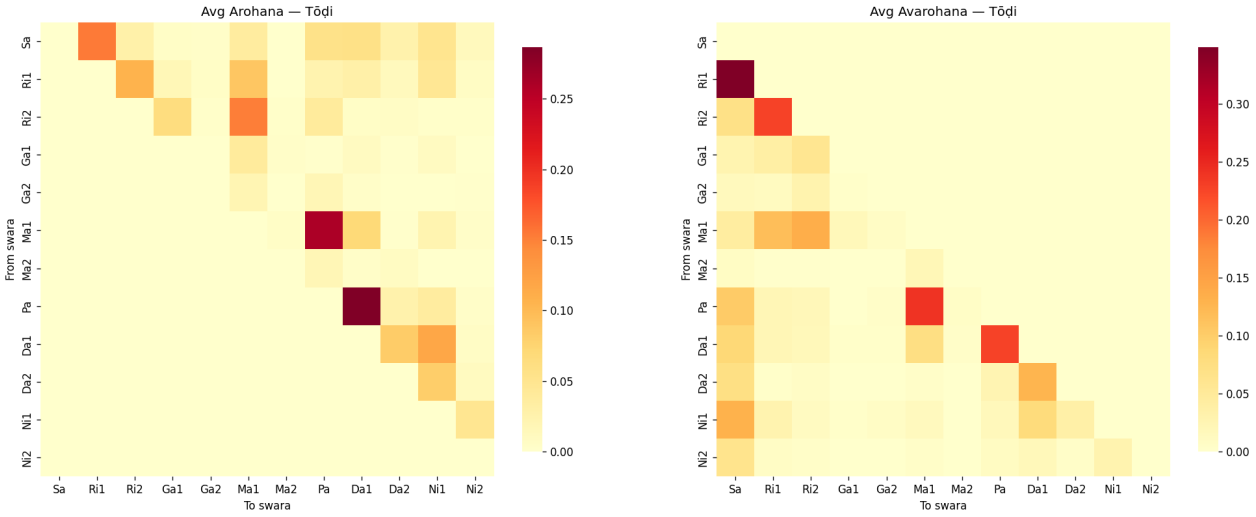
Figure 4: Swara transition matrices showing $P(\text{next} \mid \text{current})$. Tōḍi's transitions are concentrated along specific paths; Kamās shows distributed patterns typical of the Harikāmbhōji family.



(a) Surați (b) Mōhanaṁ

Figure 5: Additional transition matrices. Mōhanaṁ (pentatonic) shows a sparser 5×5 active region.

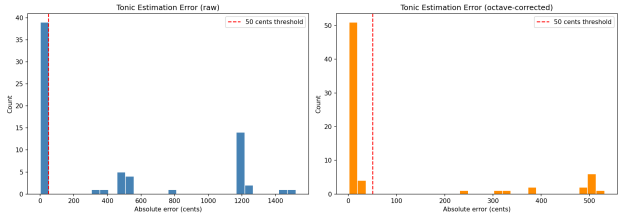
3.3 Directional Features (Ārohana / Avarohana)



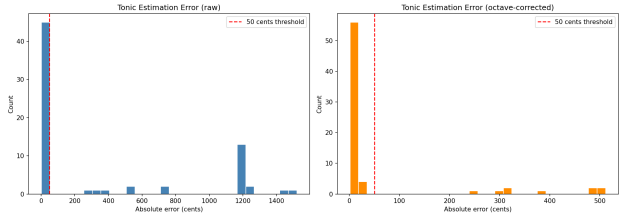
(a) Tōḍi ascending (ārohana) (b) Tōḍi descending (avarohana)

Figure 6: Directional PCD for Tōḍi. Tested under H6; directional encoding did not improve classification because gamakas blur the ascending/descending distinction in performed Carnatic music.

4 Tonic Estimation: All Methods Compared



(a) Sa-mass scoring (79.7% within 50 cents)



(b) Harmonic scoring (87.0% within 50 cents)

Figure 7: Tonic error histograms. Harmonic scoring produces a tighter distribution around zero cents error by exploiting the tānpura's Sa+Pa drone structure.

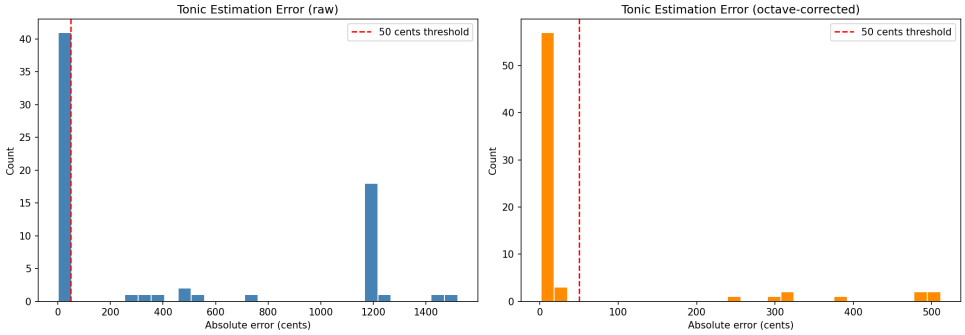
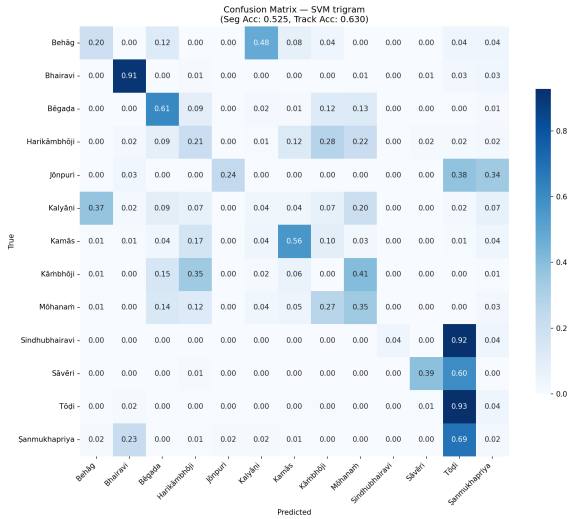


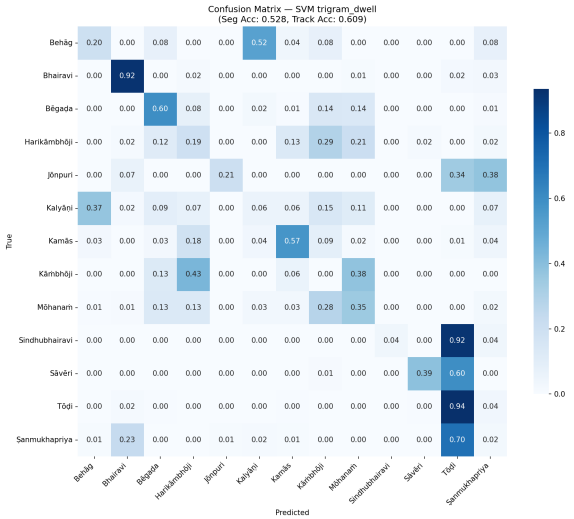
Figure 8: Smoothed harmonic scoring ($\sigma = 2$ bins): reduces narrow-peak sensitivity but does not improve overall accuracy. More octave errors than unsmoothed harmonic.

5 Classification: All Confusion Matrices

5.1 Primary Feature Sets

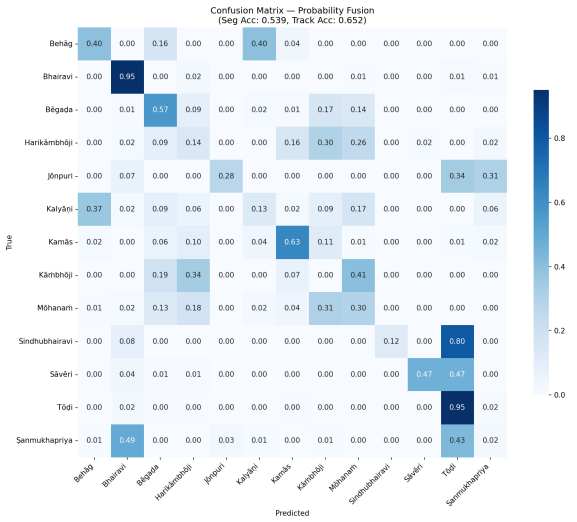


(a) Trigram (200d) — 63.0% track



(b) Trigram + Dwell (216d) — 60.9% track

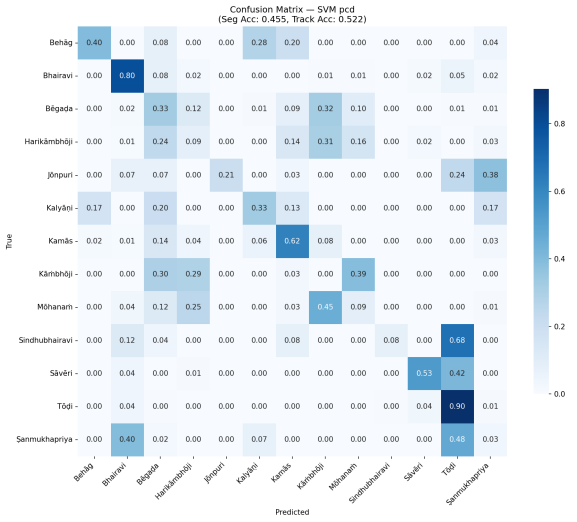
Figure 9: Confusion matrices for trigram-based feature sets. Both show within-family confusion in the Harikāmbhōji cluster and clean separation of Tōḍi.



(a) Probability fusion — 65.2% track (b) BiLSTM — 45.3% track (GT tonic)

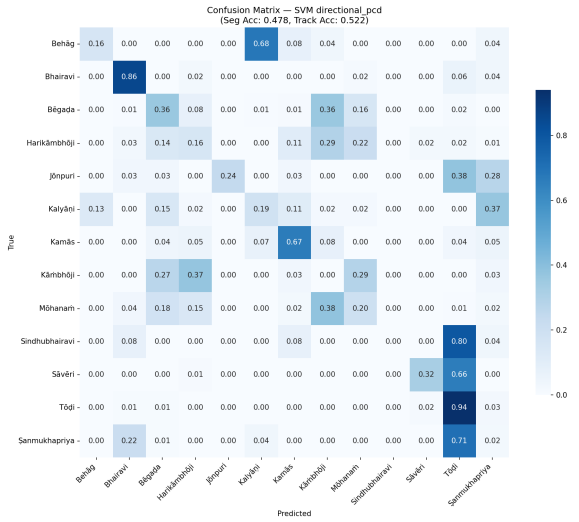
Figure 10: Fusion and deep learning baseline. The BiLSTM suffers from data starvation (2,107 segments, best single fold: 74.17%).

5.2 Additional Feature Variants



(a) PCD (120d) — 52.2% track (b) PCD + Trigram (320d) — 56.5% track

Figure 11: PCD-based confusion matrices. Adding PCD to trigrams does not improve over trigrams alone (H3: cognitive parsimony).

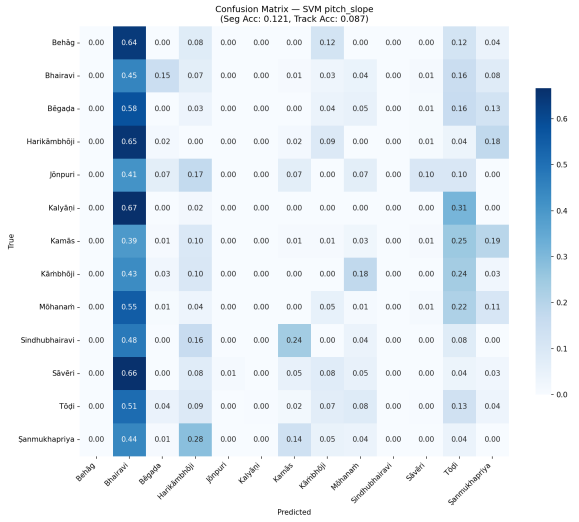


(a) Directional PCD (408d) — 52.2% track (b) Transition matrix (144d)

Figure 12: Directional (H6, does not improve) and transition matrix confusion matrices.

6 Negative Results and Failed Experiments

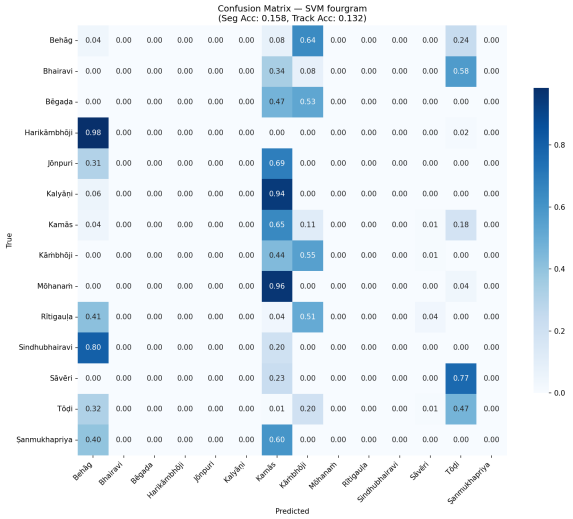
6.1 Gamaka Features



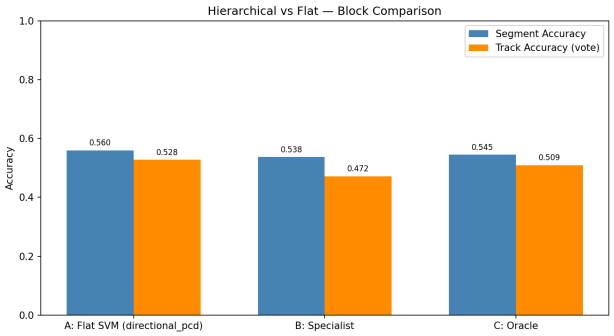
(a) Pitch slope features (b) Combined gamaka features

Figure 13: Gamaka experiment. Continuous pitch contour features do not improve classification — pitch slope distributions are pan-rāga (all Carnatic music uses heavy ornamentation) rather than rāga-specific.

6.2 4-gram and Hierarchical



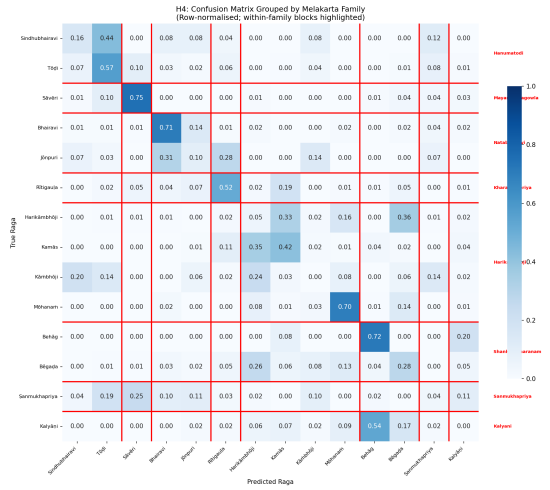
(a) 4-gram (15.8% — fatally sparse at 12^4 dims)



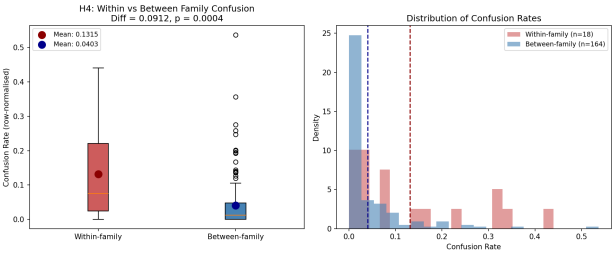
(b) Hierarchical vs flat comparison

Figure 14: Two failed experiments. 4-grams: $12^4 = 20{,}736$ possible patterns create fatal sparsity. Hierarchical: flat baseline matches or exceeds hierarchy because 60.7% of errors are within-family (where L1 routing cannot help).

6.3 H4: Melakarta Family Confusion Analysis



(a) Confusion matrix annotated by melakarta family



(b) Within vs between-family confusion ($p = 0.0004$)

Figure 15: H4 confirmed: within-family confusion is 3.27× higher than between-family under permutation test (10,000 iterations).

7 Hindustani Cross-Tradition Pilot

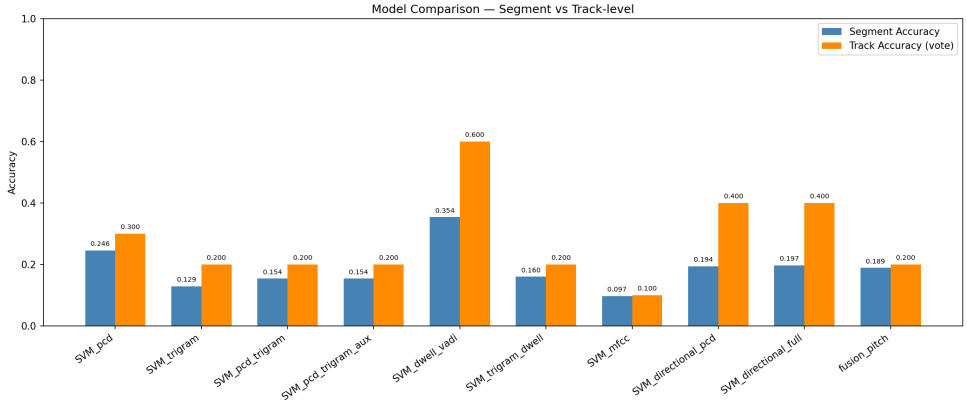
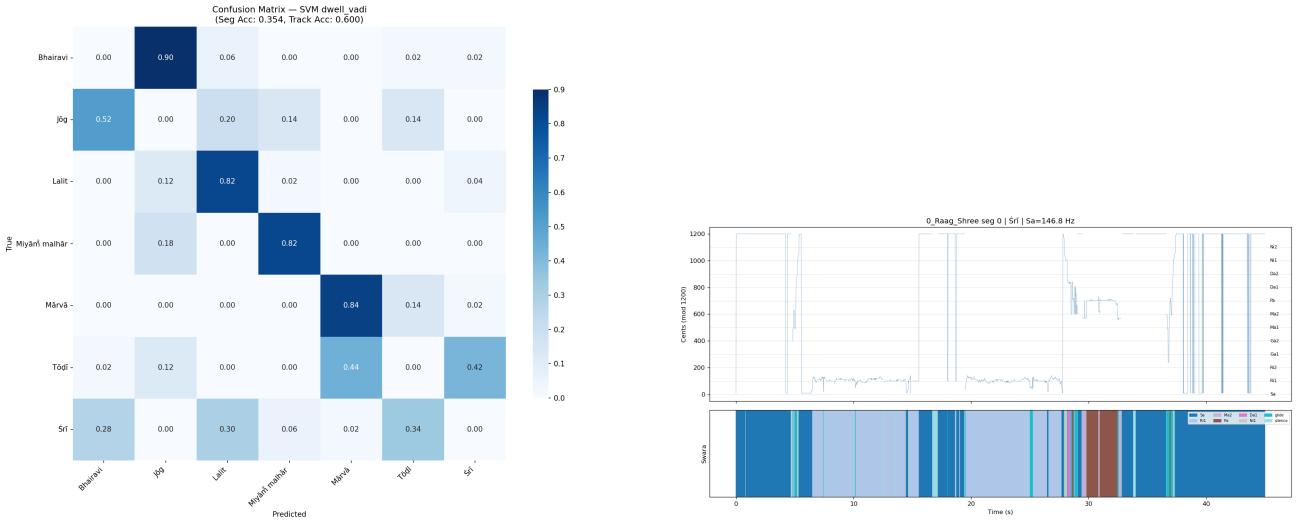


Figure 16: Hindustani model comparison (10 tracks). Feature-importance flip: dwell_vādi dominates while trigrams collapse — consistent with Hindustani music being emphasis-driven rather than phrase-driven.



(a) Hindustani dwell_vādi

(b) Hindustani swara mapping: Raag Shree

Figure 17: Hindustani results: dwell_vādi confusion matrix and swara mapping, demonstrating the pipeline operating on a different tradition.